

Mariia (Mary) Feofanova

Immigration status: London, SW visa

Open to remote work: yes

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EXPERIENCE

Thought Machine

London, UK

- *SWE, Ledger, Postings Processing team*

Aug. 2023 - Now

[GoLang](#) [Python](#) [Kubernetes](#) [Grafana](#) [GeminiCLI](#)

- Developing and maintaining core services responsible for processing and storing financial transactions, balancing high performance with strict consistency.
- Led a team of four engineers to make certain types of transactions traffic more efficient. It resulted in 2x gain in the throughput, proven by internal performance testing.
- Designed a reconciliation tool for the company that helped finding multiple financial consistency bugs. Led a team of two engineers to implement and productionise the tool. After my work proved being useful, it is officially integrated with the product and runs on all environments by default.

Google LLC

London, UK

- *SRE, Cloudnet Control SRE*

Oct. 2022 - Apr. 2023

[GoLang](#) [Borg](#) [Monitoring](#)

- Led a project to migrate a monitoring infrastructure to a new framework. This infrastructure measures ssh connectivity to VMs, and has detected global incidents. The migration enabled the monitoring team to turn off their legacy framework.
- Participated in oncall interrupts shifts.

Tarantool, Mail.ru Group

Moscow, Russia

- *SWE, Tarantool Core team*

Sept. 2020 - Apr. 2021

[C](#) [C++](#) [Lua](#)

- Was working on the core of [Tarantool](#) - an open-source database used by the biggest tech companies in Russia (Mail.ru, Badoo, Mastercard, etc.).
- Improved functionality of indexes: added binary type support for bitset index, implemented option to exclude null values.
- Introduced interactive transactions over the iproto protocol so that now transactions don't need to be sent in one request.

JetBrains s.r.o

Moscow, Russia

- *SWE, RubyMine team*

Jul. 2018 - Feb. 2019

[Java](#) [Ruby](#) [Ruby on Rails](#)

- Improved refactoring system for RubyMine IDE: developed type specification for [extract parameter](#), added new extraction features, implemented prompt that analyses where to insert missing "end" keyword.

INTERNSHIPS

Google LLC

Remote

- *SWE Internship, Infra Bigtable SRE*

July 2021 - Sept. 2021

[GoLang](#) [Borg](#)

- I designed, implemented, and tested a prober to monitor the Bigtable backup system. The prober was written in Go, and ran on Google's cluster management system. It ran continuously to detect backup problems before customers.

JetBrains s.r.o

Remote

- *Research Internship, Programming Languages and Tools Lab*

July 2020 - Aug. 2020

C LLVM

- Modified [GenMC](#) model checker (tool for verifying concurrent programs under weak memory models) to test production code: added intrinsics support, sped up allocation instrumentation by more than 2 times.
- Tested [mimalloc](#) (Microsoft allocator) using GenMC, designed and implemented 6 tests. [Collaborated](#) with mimalloc maintainer [Daan Leijen](#) to make mimalloc data-race-free, which resulted in confirming and fixing 6 data races (e.g., [wrong memory order](#)). Additionally found 2 tricky [bugs](#) which could not be confirmed by other tools.

Google LLC

London, UK

- *SWE Internship, Android App Compat team*

July 2019 - Sept. 2019

GoLang Java Android Protobuf SQL

- Designed and implemented presubmit checker for Android API repository: it analyses API usage statistics and prevents developers from submitting changes that break popular APIs.

PROJECTS

[Model checker for testing concurrent programs](#) [Bachelor thesis]

Moscow, Russia

- C++ TLA+ +Cal

Nov. 2019 - May 2020

- Examined pitfalls of two common concurrency testing techniques (fault injection and verification using +Cal) and created my own solution: model checker that avoids these pitfalls.
- Verified the tool on non-trivial examples. In one of them, my model checker finds complex A-B-A bug 16 times faster than +Cal.
- Made model checker easy to use and understand: it provides an API to create expressive tests and shows steps to reproduce a bug.

TEACHING EXPERIENCE

Moscow Institute of Physics and Technology

Moscow, Russia

- *Teaching Assistant*

Feb. 2019 - June 2021

C++ x86 Assembly

- **Theory and Practice of Concurrent Computing:** holding seminars and reviewing homework. Learned about internals of modern concurrency concepts from industry, designed new tests for assignments, and contributed to the testing framework by writing model checker as a part of my Bachelor thesis.

ACHIEVEMENTS

Hackathon winner - [Photo Lab Live Challenge: Go Viral or Go Home](#)

Moscow, Russia

- Python web.py Heroku

Nov. 2018

Developed backend for photo filter that changes the style of clothes. Having only 48 hours was able to quickly learn python web framework and to deploy MVP by collaborating with two teammates.

EDUCATION

Moscow Institute of Physics and Technology

Moscow, Russia

- *Bachelor of Applied Mathematics and Computer Science*

Sept. 2016 - Aug. 2020

Description: Joint degree with the [Yandex School of Data Analysis](#): the most prestigious and selective CS private study program in Russia (~3% acceptance rate - only 129 students admitted out of ~4300). It is offered by Yandex, one of the most technologically advanced Russian software companies.

TECHNICAL SKILLS

- **Programming Languages:** C++, C, GoLang, Java, Python, SQL, TLA+ / +Cal
- **Knowledge:** Algorithms, Data Structures, Databases, Concurrent Computing, Distributed Systems